



DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING

UNIVERSITY OF DHAKA

**Title: Interpolation Techniques: Lagrange
Interpolation & Newton's Divided Difference
Method**

CSE 3212: NUMERICAL ANALYSIS LAB
BATCH: 29/3RD YEAR 2ND SEMESTER 2025

1 Objective(s)

- To understand the concept and need of interpolation for the estimation of unknown values.
- Implement Lagrange Interpolation to approximate values of a function using tabulated data.
- Implement Newton's Divided Difference Interpolation and construct the forward interpolation polynomial.
- Compare efficiency, computational complexity, and suitability of both interpolation techniques.

2 Background Theory

Interpolation is a numerical method used to estimate the value of a function between two known values. It constructs a polynomial that passes through a given set of data points. “The following two interpolation approaches are commonly used in scientific computing and data fitting. Each constructs a polynomial passing through all given data points, but differs in formulation and computational cost.

2.1 The Lagrange Polynomial

Lagrange interpolation constructs an interpolating polynomial of degree n for given data points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$.

$$P_n(x) = \sum_{i=0}^n y_i L_i(x)$$

where

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}$$

Characteristics

- Simple to understand.
- No need to construct a difference table.
- Requires recomputation if a new data point is added.

2.2 Newton's Divided Difference Interpolation

Newton's interpolation uses a divided difference table to form an interpolation polynomial.

Divided Difference Table Formula

$$f[x_i, x_{i+1}] = \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i}$$

General Divided Difference

$$f[x_i, x_{i+1}, \dots, x_{i+k}] = \frac{f[x_{i+1}, \dots, x_{i+k}] - f[x_i, \dots, x_{i+k-1}]}{x_{i+k} - x_i}$$

Newton Polynomial

$$P_n(x) = f[x_0] + (x - x_0)f[x_0, x_1] + (x - x_0)(x - x_1)f[x_0, x_1, x_2] + \dots$$

Characteristics

- Efficient for incremental addition of data points.
- Uses a difference table — easier for computation.
- Suitable for numerical algorithms.

2.3 Difference Between Lagrange and Newton Interpolation

Feature	Lagrange Interpolation	Newton's Divided Difference
Complexity	High for new data addition	Efficient for adding new points
Approach	Direct formula	Table-based recursive computation
Programming Ease	Simple	Slightly complex but scalable
Best Use	Small data sets	Large data sets & incremental updates

3 Sample Data for Demonstration

This section demonstrates manual calculation steps for both methods using a small cubic dataset to ensure conceptual understanding before coding

x	$f(x)$
1	1
2	8
3	27

Estimate $f(2.5)$.

3.1 Lagrange Interpolation: Step-by-Step Calculation

Given points: (1, 1), (2, 8), (3, 27)

$$L_0(x) = \frac{(x-2)(x-3)}{(1-2)(1-3)} = \frac{(x-2)(x-3)}{2}$$

$$L_1(x) = \frac{(x-1)(x-3)}{(2-1)(2-3)} = -(x-1)(x-3)$$

$$L_2(x) = \frac{(x-1)(x-2)}{(3-1)(3-2)} = \frac{(x-1)(x-2)}{2}$$

Now form:

$$P(x) = 1 \cdot L_0(x) + 8 \cdot L_1(x) + 27 \cdot L_2(x)$$

Substitute $x = 2.5$ to compute the value.

3.2 Newton's Divided Difference: Step-by-Step Calculation

Divided Difference Table

x	$f[x]$	$f[x_0, x_1]$	$f[x_0, x_1, x_2]$
1	1	$\frac{8-1}{2-1} = 7$	$\frac{27-1}{3-1} = 13$
2	8	$\frac{27-8}{3-2} = 19$	
3	27		

Newton Polynomial

$$P(x) = 1 + (x-1)(7) + (x-1)(x-2)(13)$$

Evaluate at $x = 2.5$.

4 Algorithms

4.1 Algorithm: Lagrange Interpolation

1. Read n and data points $(x[i], y[i])$.
2. Read the value of x for which $f(x)$ is required.
3. Initialize **result** = 0.

4. For $i = 0$ to $n - 1$:

(a) Compute $L_i = 1$.

(b) For $j = 0$ to $n - 1$, $j \neq i$:

$$L_i = L_i \times \frac{x - x[j]}{x[i] - x[j]}$$

(c) `result = result + L_i * y[i]`.

5. Display result.

4.2 Algorithm: Newton's Divided Difference

1. Read n and input data points $(x[i], y[i])$.

2. Construct divided difference table.

3. Initialize `result = f[x0]`.

4. For $k = 1$ to $n - 1$:

(a) `term = f[x0, x1, ..., xk]`.

(b) For $j = 0$ to $k - 1$:

$$\text{term} = \text{term} * (x - x[j])$$

(c) `result = result + term`.

5. Display result.

5 Lab Tasks (Please implement yourself and show the output to the instructor)

In this lab, students will work with a real-life interpolation problem related to Bangladesh or any global issue. Instead of using only predefined datasets, students must collect or generate their own data from practical observations, online resources, government databases, IoT sensors, weather reports, transportation systems, economics, agriculture, healthcare, energy systems, or environmental monitoring.

5.1 Problem-Based Real-Life Data Collection

Choose **ONE** real-world scenario from Table 1, where interpolation can be used to estimate unknown values between measured data points. Students must follow the corresponding independent and dependent variable structure associated with the selected domain.

For the selected scenario, students must collect or generate:

- At least 15–20 real or simulated data points
- One independent variable x
- One dependent variable $f(x)$

5.2 Task 1 — Lagrange Interpolation for Real-World Data Estimation

Using the real-world dataset collected/generated from the selected scenario in Table 1, perform the following tasks using the **Lagrange Interpolation Method**.

1. **Dataset preparation and target selection**

- (a) Select one unknown/intermediate value $x = x^*$ within the observed data range where the corresponding value $f(x^*)$ is not directly available.
- (b) Choose suitable neighboring data points around x^* for interpolation. Clearly justify your node selection strategy.

Table 1: Real-World Domains, Example Problems, and Corresponding Dataset Structures for Interpolation

Domain	Example Problem	Independent Variable (x)	Dependent Variable ($f(x)$)
Climate & Weather	Predict temperature/rainfall between measurement times	Time	Temperature/Rainfall
Traffic System	Estimate traffic density between two observation points	Time	Vehicle count
Fuel/Energy Crisis	Estimate fuel demand or pump queue length	Day/Time	Fuel demand or queue length
Agriculture	Predict crop moisture or soil temperature	Time	Soil moisture/temperature
River Monitoring	Estimate water level between sensor locations	Distance	Water depth
Air Pollution	Estimate AQI between monitoring stations	Location	AQI
Healthcare	Estimate patient body temperature/glucose trends	Time	Temperature/Glucose level
Transportation	Estimate bus/train arrival times	Distance/Time	Arrival time
Internet & Networking	Estimate bandwidth variation	Time	Bandwidth usage
Economics	Predict commodity prices between dates	Date/Time	Commodity price
Smart City	Estimate electricity usage in urban regions	Time	Electricity consumption

2. Construct and evaluate low-order interpolants

- Construct a *second-order* (quadratic) Lagrange interpolating polynomial $P_2(x)$ using three nearby data points.
- Evaluate the estimated value:
$$P_2(x^*)$$
- Interpret the estimated result in the context of your selected real-world problem.

3. Increase polynomial degree and monitor convergence

- Construct higher-order interpolating polynomials:
$$P_3(x), P_4(x), \dots$$

using additional nearby data points.

- For each interpolating polynomial $P_k(x)$, evaluate:
$$P_k(x^*)$$

- Compute the convergence difference:
$$\Delta_k = |P_k(x^*) - P_{k-1}(x^*)|$$

- Analyze how interpolation accuracy and stability change as the polynomial degree increases.

4. Visualization and analysis Plot Original data points, Interpolation curves for different polynomial degrees and Estimated/interpolated points

5. Deliverables for Task 1

- Collected/generated dataset
- The Lagrange basis functions used (symbolic or numeric).
- The expanded polynomial(s) (coefficients to at least 6 significant digits).
- Values $P_k(x^*)$ for each degree used and the Δ_k sequence.
- Provide tables of $P_k(x^*)$, convergence differences Δ_k , and a plot of $P_k(x)$ for all degrees used
- Graphical plots and analysis

5.3 Task 2 — Newton's Divided Difference Interpolation for Real-World Data Estimation

Using the same real-world dataset, perform the following tasks using the **Newton's Divided Difference Interpolation Method**.

1. **Node selection and ordering**

- (a) Select and order interpolation nodes appropriately around the target value $x = x^*$.
- (b) Explain the node ordering strategy used to improve interpolation accuracy.

2. **Construct divided difference table**

Compute the complete divided difference table for the selected nodes, including:

$$f[x_i], \quad f[x_i, x_{i+1}], \quad f[x_i, x_{i+1}, x_{i+2}], \dots$$

Use sufficient numerical precision (at least 6 significant digits).

3. **Construct Newton interpolation polynomials**

- (a) Construct Newton interpolation polynomials of different degrees:

$$N_2(x), N_3(x), N_4(x), \dots$$

- (b) Evaluate:

$$N_k(x^*)$$

for each polynomial degree.

- (c) Compute convergence differences:

$$\Delta_k = |N_k(x^*) - N_{k-1}(x^*)|$$

- (d) Compare the convergence behavior with the Lagrange interpolation method.

4. **Visualization and comparative analysis** Plot Newton interpolation curves, Original data points and Estimated points for all polynomial degrees.

Deliverables for Task 2

- Complete divided difference table
- Newton interpolation polynomial(s)
- Estimated values $N_k(x^*)$
- Convergence differences Δ_k
- Comparative analysis with Lagrange interpolation
- Graphical plots and discussion

6 Lab Report: Spline Interpolation

In many real-world applications, high-degree polynomial interpolation methods may produce oscillations and unstable approximations. Spline interpolation addresses this issue by constructing smooth piecewise polynomials between consecutive data points.

In this lab report, students will use the **same real-world dataset and problem domain** used in the lab tasks of interpolation (Lagrange and Newton's Divided Difference methods). Using the collected/generated dataset, estimate unknown values between measured data points using:

- Linear Spline Interpolation
- Quadratic Spline Interpolation
- Cubic Spline Interpolation

6.1 Tasks

1. Construct linear, quadratic, and cubic spline interpolation functions for the dataset.
2. Estimate the unknown/intermediate value:

$$f(x^*)$$

where x^* is the same target interpolation point used previously.

3. Plot:
 - Original data points
 - Linear spline curve
 - Quadratic spline curve
 - Cubic spline curve
4. Compare the interpolation methods based on:
 - Smoothness
 - Stability
 - Approximation quality
 - Computational complexity

6.2 Deliverables

Students must submit:

- Dataset used for interpolation
- Spline interpolation equations
- Estimated values
- Graphical plots
- Comparative analysis and discussion